

CHIA-HAO CHANG-YANG

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Department of Biological Sciences
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Identifier

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EDUCATION

- **Ph.D. 2013.** Institute of Ecology and Evolutionary Biology. National Taiwan University, Taiwan.
Long-term Dynamics of Seed Rain and Seedling in a Subtropical Rain Forest, Fushan, Taiwan.
Committee: Chang-Fu Hsieh (Chair), Kuo-Jung Chao, Yau-Lun Kuo, Yi-Ching Lin, I-Fang Sun
- **M.S. 2004.** Institute of Ecology and Evolutionary Biology. National Taiwan University, Taiwan.
The Seed Rain in a Subtropical Rain Forest, Fushan, Northern Taiwan.
Committee: Chang-Fu Hsieh (Chair), Wen-Liang Chiou, Yau-Lun Kuo, I-Fang Sun
- **B.S. 2002.** Department of Botany. National Taiwan University, Taiwan.

RESEARCH / WORK EXPERIENCE

- **Sep. 2016–Jul 2018.** Postdoctoral Researcher, National Dong Hwa University.
Effect of Global Warming on Taiwan Forest Ecosystems
PI: I-Fang Sun
- **Aug. 2105–Aug. 2016.** Postdoctoral Researcher, Smithsonian Environmental Research Center.
Trait-mediated Assembly Mechanisms along an Environmental Stress Gradient
PI: Sean M. McMahon
- **Aug. 2014–July 2015.** Postdoctoral Researcher, Tunghai University.
Effects of Deer Browsing on Species Composition in a Karst Forest, Kenting
PI: Yi-Ching Lin
- **Feb. 2013–July 2014.** Postdoctoral Researcher, National Dong Hwa University.
Competition, Phylogeny, and Functional Traits in a Subtropical Rain Forest
PI: I-Fang Sun
- **Aug. 2007–Jan. 2013.** Dissertation Research, National Taiwan University.
Long-term Dynamics of Seed Rain and Seedling in a Subtropical Rain Forest, Fushan, Taiwan
Advisor: Chang-Fu Hsieh
- **Jan. 2007–July 2007.** Research Assistant, Taiwan Forestry Research Institute.
Long-term Dynamics of the Fushan Broad-leaved Forest
PIs: Tzer-Tong Lin & Sheng-Hsin Su
- **Jan. 2006–Dec. 2006.** Research Assistant, Tunghai University
Seed Rain and Seedling Demography in a Subtropical Rain Forest, Fushan, Taiwan
PI: I-Fang Sun
- **July 2004 – Dec. 2005.** Research Assistant, National Taiwan University

First Census of the Fushan 25-ha Long-term Plot and Phenology Study

PI: Chang-Fu Hsieh

- **June 2002–June 2004.** Master Thesis Research, National Taiwan University.
The Seed Rain in a Subtropical Rain Forest, Fushan, Northern Taiwan
Advisor: Chang-Fu Hsieh
- **May 2001–Apr. 2002.** Undergraduate Independent Research, National Taiwan University.
*Population Structure and Short-term Dynamics of *Drypetes karapinensis* in Nanjenshan Area*
Advisor: Chang-Fu Hsieh

PUBLICATIONS

- Johnson, D. J., Needham, J., Xu, C., Massoud, E. C., Davies, S. J., Anderson-Teixeira, K. J., Bunyavejchewin, S., Chambers, J. Q., **Chang-Yang, C.-H.**, Chiang, J.-M., Chuyong, G. B., Condit, R., Cordell, S., Fletcher, C., Giardina, C. P., Giambelluca, T. W., Gunatilleke, N., Gunatilleke, S., Hsieh, C.-F., Hubbell, S., Inman-Narahari, F., Kassim, A. R., Katabuchi, M., Kenfack, D., Litton, C. M., Lum, S., Mohamad, M., Nasardin, M., Ong, P. S., Ostertag, R., Sack, L., Swenson, N. G., Sun, I. F., Tan, S., Thomas, D. W., Thompson, J., Umaña, M. N., Uriarte, M., Valencia, R., Yap, S., Zimmerman, J., McDowell, N. G. & McMahon, S. M. (2018) Climate sensitive size-dependent survival in tropical trees. *Nature Ecology & Evolution*, **2**, 1436-1442.
- Hogan, J., Zimmerman, J., Thompson, J., Uriarte, M., Swenson, N., Condit, R., Hubbell, S., Johnson, D., Sun, I.-F., **Chang-Yang, C.-H.**, Su, S.-H., Ong, P., Rodriguez, L., Monoy, C., Yap, S. & Davies, S. (2018) The Frequency of Cyclonic Wind Storms Shapes Tropical Forest Dynamism and Functional Trait Dispersion. *Forests*, **9**, 404.
- Needham, J., Merow, C., **Chang-Yang, C.-H.**, Caswell, H. & McMahon, S. M. (2018) Inferring forest fate from demographic data: from vital rates to population dynamic models. *Proceedings of the Royal Society B: Biological Sciences*, **285**, 20172050.
- Usinowicz, J., **Chang-Yang, C.-H.**, Chen, Y.-Y., Clark, J.S., Fletcher, C., Garwood, N.C., Hao, Z., Johnstone, J., Lin, Y., Metz, M.R., Masaki, T., Nakashizuka, T., Sun, I.-F., Valencia, R., Wang, Y., Zimmerman, J.K., Ives, A.R. & Wright, S.J. (2017) Temporal coexistence mechanisms contribute to the latitudinal gradient in forest diversity. *Nature*, **550**, 105-108.
- Herrmann, V.M.D., McMahon, S.M., Detto M., Lutz, J.A., Davies, S.J., **Chang-Yang, C.-H.**, & Anderson-Teixeira, K.J. (2016) Tree circumference dynamics in four forests characterized using automated dendrometer. *PLoS ONE*, **11**, e0169020.
- Chang-Yang, C.-H.**, Sun, I.-F., Tsai, C.-H., Lu, C.-L. & Hsieh, C.-F. (2016) ENSO and frost co-determine decade-long temporal variation in flower and seed production in a subtropical rain forest. *Journal of Ecology*, **104**, 44-54.
- Chang-Yang, C.-H.**, Su, S.-H., Wang, H.-H., Lin, T.-T. & Hsieh, C.-F. (2015) Topography and canopy openness influence spatial variations in mortality, recruitment, and growth of a subtropical tree community. *Taiwan Journal of Forest Science*, **30**, 259-269.
- Zhong, L., **Chang-Yang, C.-H.**, Lu, P., Gu, X., Lei, Z., Cai, Y., Zheng, F., Sun, I.-F. & Yu, M. (2015) Community structure and species composition of the secondary evergreen broad-leaved forest: the analyses for a 9-ha forest dynamics plot in Wuyanling Nature Reserve, Zhejiang Province, East China. *Biodiversity Science*, **23**, 619-629.
- Hu, X., **Chang-Yang, C.-H.**, Mi, X., Du, Y. & Chang, Z. (2015) The influence of climate, phylogeny, and functional traits on flowering phenology in a subtropical evergreen broad-leave forest, east China. *Biodiversity Science*, **23**, 601-609.
- Chang-Yang, C.-H.**, Lu, C.-L., Sun, I.-F. & Hsieh, C.-F. (2013) Long-term seedling dynamics of tree species in a subtropical rain forest, Taiwan. *Taiwania*, **58**, 35-43.
- Chang-Yang, C.-H.**, Lu, C.-L., Sun, I.-F. & Hsieh, C.-F. (2013) Flowering and fruiting patterns in a subtropical rain forest, Taiwan. *Biotropica*, **45**, 165-174.

- Su, S.-H., Hsieh, C.-F., **Chang-Yang, C.-H.**, Lu, C.-L. & Guan, B.-T. (2010) Micro-topographic differentiation of the tree species composition in a subtropical submontane rainforest in northeastern Taiwan. *Taiwan Journal of Forestry Science*, **25**, 63-80.
- Su, S.-H., **Chang-Yang, C.-H.**, Lu, C.-L., Tsui, C.-C., Lin, T.-T., Lin, C.-L., Chiou, W.-L., Kuan, L.-H., Chen, Z.-S. & Hsieh, C.-F. (2007) *Fushan Subtropical Forest Dynamics Plot: Tree Species Characteristics and Distribution Patterns*. Taiwan Forestry Research Institute, Taipei, Taiwan.

HONOR, GRANTS, & FELLOWSHIPS

- **2017.** Postdoctoral Research Fellowship. Ministry of Science and Technology, Taiwan.
- **2016.** Postdoctoral Research Fellowship. Ministry of Science and Technology, Taiwan.
- **2015.** Postdoctoral Research Abroad Program Fellowship. Ministry of Science and Technology, Taiwan.
- **2014.** Postdoctoral Research Fellowship. Ministry of Science and Technology, Taiwan.
- **2013.** Science and Technology Program Grant. Taiwan Forestry Bureau, Taiwan. Topic: *Vulnerability and risk assessments for biodiversity to climate change: Climatic effects on plant phenology in a subtropical rain forest.*
- **2010.** International Conference Traveling Grant. Ministry of Education, Taiwan
- **2004.** Graduate Student Poster Presentation Excellent Award. Institute of Ecology and Evolutionary Biology, National Taiwan University, Taiwan.
- **2001.** Undergraduate Independent Research Grant. National Science Council, Taiwan.

TRAINING WORKSHOPS

- **July 2015.** Smithsonian Tropical Research Institute, Smithsonian Institute.
Center for Tropical Forest Science-Smithsonian Institute Global Earth Observatory (CTFS-SIGEO) Analytical Workshop, Panama.
Mentors: S. Joseph Wright & Liza S. Comita
- **July 2014.** Xishungbanna Tropical Botanical Garden, Chinese Academy of Science.
Center for Tropical Forest Science-Smithsonian Institute Global Earth Observatory (CTFS-SIGEO) Analytical Workshop, China.
Mentors: Sean M. McMahon, Sabrina E. Russo, & Stuart J. Davies
- **July 2013.** Smithsonian Conservation Biology Institute, Smithsonian Institute.
Center for Tropical Forest Science-Smithsonian Institute Global Earth Observatory (CTFS-SIGEO) Analytical Workshop, USA.
Mentors: S. Joseph Wright & Helene C. Muller-Landau
- **May 2013.** National Dong Hwa University.
International Workshop on Ecological Statistics, Taiwan.
Instructor: Pierre Legendre
- **June 2012.** National Dong Hwa University.
Workshop on Ecological Statistical Modelling, Taiwan.
Instructor: Jesse R. Lasky
- **May 2012.** National Dong Hwa University.
Phylogenetic and Functional Trait Analyses of Communities Workshop, Taiwan.
Instructor: Nathan G. Swenson
- **Apr. 2009.** Tunghai University & Taiwan Forestry Research Institute.
Workshop on Likelihood Methods, Taiwan.
Instructor: Charles D. Canham

- **Apr. 2008.** Tunghai University & Smithsonian Tropical Research Institute.
Workshop on Plant Functional Traits and Carbon Budget, Taiwan
Instructors: S. Joseph Wright & Helene C. Muller-Landau
- **Aug. 2007.** Research Center for Biodiversity, Academia Sinica.
Analytical Workshop on Quantitative Ecology, Taiwan.
Instructor: Pierre Legendre, Daniel A. Griffith, & Fangliang He
- **June 2005.** Smithsonian Tropical Research Institute, Smithsonian Institute.
Center for Tropical Forest Science Analytical Workshop, Panama.
Instructor: Richard Condit

PRESENTATIONS & INVITED SEMINARS

- 2019. Chang-Yang, C.-H.** *Climate fluctuations, extreme weather events, and forest dynamics: Lessons from the long-term ecological monitoring in Fushan.* Department of Life Science, National Taiwan University. Taipei, Taiwan. Invited seminar.
- 2018. Chang-Yang, C.-H.** *Effects of ENSO and typhoons on the flower and seed production in a subtropical rain forest.* Global Change Research Center, National Taiwan University. Taipei, Taiwan. Invited seminar.
- 2018. Chang-Yang, C.-H.** *Closing the life cycle of forest trees: Seedling-to-sapling transition in a subtropical rain forest.* The 6th Taiwan-Japan Ecology Workshop. Tainan, Taiwan. Invited oral presentation.
- 2018. Chang-Yang, C.-H.,** Needham J., Sun I.-F., Lu C.-L., Hsieh C.-F., & McMahon S. M. *Diversity Takes Time: Temporal Dynamics of Seedling Demography in a Subtropical Rain Forest.* Association for Tropical Biology and Conservation 55th Annual Meeting. Kuching, Malaysia. Contributed oral presentation.
- 2017. Chang-Yang, C.-H.** *Closing the life cycle of forest trees: Seedling-to-sapling transition in a subtropical rain forest.* MOST Biodiversity Division Workshop. Taichung, Taiwan. Invited oral presentation.
- 2017. Chang-Yang, C.-H.** *Climate fluctuations, extreme weather events, and forest dynamics: Lessons from the long-term ecological monitoring in Fushan.* Department of Life Science, National Taiwan Normal University. Taipei, Taiwan. Invited seminar.
- 2017. Lu, J.-W., Fu, Y.-H., Chang-Yang, C.-H., Wu, S.-H. & Lin, Y.-C.** Effects of ungulate herbivory on seedling communities of woody plants in the Kenting karst forest. Symposium on Sustainable Development of Forest Resources. Ilan, Taiwan. Contributed poster presentation.
- 2017. Chang-Yang, C.-H.** *Climate fluctuations, extreme weather events, and forest dynamics: Lessons from the long-term ecological monitoring in Fushan.* Department of Biological Sciences, National Sun Yat-sen University. Kaohsiung, Taiwan. Invited seminar.
- 2016. Chang-Yang, C.-H.** *Seedling-to-sapling transition in a subtropical rain forest.* The 10th Cross-Strait Symposium on Forest Dynamics Plots. Beijing, China. Contributed oral presentation.
- 2015. Chang-Yang, C.-H.,** Sun, I.-F., Lu, C.-L. & Hsieh, C.-F. *Spatial and temporal variation in seedling recruitment of nine tree species in a subtropical rain forest.* Ecological Society of America 100th Annual Meeting. Baltimore, USA. Contributed poster presentation.
- 2015. Chang-Yang, C.-H.** *Effects of ENSO and spring frost on flower and seed production in a subtropical rain forest.* Department of Biological Resources, National Chiayi University. Chiayi, Taiwan. Invited seminar.
- 2014. Chang-Yang, C.-H.,** Sun, I.-F., Lu, C.-L. & Hsieh, C.-F. *The effects of ENSO and spring frost on flower production in a subtropical rain forest.* The 4th Taiwan-Japan Ecology Workshop. Hualien, Taiwan. Invited oral presentation.
- 2014. Chang-Yang, C.-H.** *Long-term dynamics of seed rain and seedling in a subtropical rain forest.* Department of Life Science, Tunghai University. Taichung, Taiwan. Invited seminar.

- 2014. Chang-Yang, C.-H.** *Interannual variation in seedfall and seedling recruitment in a subtropical rain forest.* The 8th Cross-Strait Symposium on Forest Dynamics Plots. Shenyang, China. Contributed oral presentation.
- 2014. Chang-Yang, C.-H.** *Interannual variation in seedfall and seedling recruitment in a subtropical rain forest.* Center for Tropical Forest Science-Smithsonian Institute Global Earth Observatory (CTFS-SIGEO) Analytical Workshop. Xishungbanna, China. Invited seminar.
- 2014. Chang-Yang, C.-H.** *Seasonal and interannual variation in plant reproduction in a subtropical rain forest.* Taiwan Ecological Research Network Annual Meeting. Taipei, Taiwan. Invited oral presentation.
- 2013. Chang-Yang, C.-H.,** Lu, C.-L., Sun, I.-F. & Hsieh, C.-F. *Interannual variation in flower and seed production in a subtropical rain forest.* The 7th Cross-Strait Symposium on Forest Dynamics Plots. Wenzhou, China. Contributed oral presentation.
- 2013. Chang-Yang, C.-H.** *Long-term dynamics of seed rain and seedling in a subtropical rain forest.* Institute of Ecology and Evolutionary Biology, National Taiwan University. Taipei, Taiwan. Invited seminar.
- 2012. Chang-Yang, C.-H.,** Lu, C.-L., Sun, I.-F., & Hsieh, C.-F. *Flowering and fruiting phenology in a subtropical rain forest, Taiwan.* Symposium on Animal Behavior, Ecology and Global Change. Taipei, Taiwan. Invited oral presentation.
- 2010. Chang-Yang, C.-H.,** Lu, C.-L., Sun, I.-F. & Hsieh, C.-F. *Inter-specific variation in recruitment limitation in a subtropical rain forest.* The 9th National Symposium on the Conservation and Sustainable Use of Biodiversity in China. Xiamen, China. Contributed oral presentation.
- 2010. Chang-Yang, C.-H.,** Lu, C.-L., Sun, I.-F. & Hsieh, C.-F. *Synchronous flowering in a subtropical rain forest, Fushan, Taiwan.* Ecological Society of America 95th Annual Meeting. Pittsburgh, USA. Contributed oral presentation.
- 2010. Chen, Y.-C., Chang-Yang C.-H. & Lin, Y. K.** *Population dynamics and temporal partition of small mammals in Guandu salt marsh.* Symposium on Animal Behavior, Ecology and Biological Education. Changhua, Taiwan. Contributed poster presentation.
- 2009. Chang-Yang, C.-H.,** Lu, C.-L., Sun, I.-F. & Hsieh, C.-F. *Seed rain and tree seedling dynamics in a subtropical rain forest, Taiwan.* Department of Life Science, National Yang-Ming University. Taipei, Taiwan. Invited seminar.
- 2009. Chang-Yang, C.-H.,** Lu, C.-L., Sun, I.-F. & Hsieh, C.-F. *Tree seedling dynamics over seven years in a subtropical rain forest, Taiwan.* The 3rd Cross-Strait Symposium on Forest Dynamics Plots. Guangzhou, China. Invited oral presentation.
- 2008. Chang-Yang, C.-H.,** Lu, C.-L., Sun, I.-F. & Hsieh, C.-F. *Flowering patterns in a subtropical rain forest, Fushan, Taiwan.* Taiwan Ecological Research Network Annual Meeting. Taichung, Taiwan. Invited oral presentation.
- 2008. Chang-Yang, C.-H.,** Lu, C.-L., Sun, I.-F. & Hsieh, C.-F. *Flowering patterns in a subtropical rain forest, Fushan, Taiwan.* The 2nd Cross-Strait Symposium on Forest Dynamics Plots and Information Management. Hangzhou, China. Invited oral presentation.
- 2007. Chang-Yang, C.-H.,** Lu, C.-L., Sun, I.-F. & Hsieh, C.-F. *Flowering and fruiting phenology of a subtropical rain forest, Fushan, Taiwan.* The 1st Cross-Strait Symposium on Forest Dynamics Plots. Taipei, Taiwan. Invited oral presentation.
- 2007. Chang-Yang, C.-H.,** Lu, C.-L., Sun, I.-F. & Hsieh, C.-F. *Flowering and fruiting phenology of a subtropical rain forest, Fushan, Taiwan.* Symposium on Frontiers of Plant Science. Taipei, Taiwan. Invited oral presentation.
- 2006. Chang-Yang, C.-H.,** Lu, C.-L., Sun, I.-F. & Hsieh, C.-F. *The Seed rain and seedling dynamics in the Fushan subtropical rain forest.* Taiwan Ecological Research Network Annual Meeting. Taipei, Taiwan. Contributed oral presentation.
- 2006. Chang-Yang, C.-H.,** Lu, C.-L., Su, S.-H. & Hsieh, C.-F. *The seed rain and the seedling dynamics in the Fushan subtropical rain forest.* Faculty of Biomedical Science and Environmental Biology, Kaohsiung Medical University, Kaohsiung, Taiwan. Invited seminar.
- 2006. Chang-Yang, C.-H.,** Lu, C.-L. & Hsieh, C.-F. *The seed rain and the seedling dynamics in the*

Fushan subtropical rain forest. Taiwan Forest Dynamics Plot Network Symposium. Taichung, Taiwan. Invited oral presentation.

- 2006. Chang-Yang, C.-H.,** Lu, C.-L., Su, S.-H. & Hsieh, C.-F. *Regeneration strategies of dominant tree species in a subtropical rain forest in northern Taiwan*. Association for Tropical Biology and Conservation Annual Meeting. Kunming, China. Contributed poster presentation.
- 2005. Chang-Yang, C.-H.,** Lu, C.-L. & Hsieh, C.-F. *The seed rain and seedling establishment in the Fushan subtropical rain forest, Taiwan*. Ecological Society of America 90th Annual Meeting. Montral, Canada. Contributed poster presentation.
- 2005.** Lu, C.-L., **Chang-Yang, C.-H.** & Hsieh, C.-F. *The seed rain and seedling establishment of tree species in a subtropical rain forest in Fushan, northern Taiwan*. Center for Tropical Forest Science Symposium. Panama City, Panama. Contributed poster presentation.
- 2004. Chang-Yang, C.-H.** & Hsieh, C.-F. *The seed rain in a subtropical rain forest, Fushan, northern Taiwan*. Center for Tropical Forest Science Symposium. Taipei, Taiwan. Contributed poster presentation.

TEACHING & MENTORING EXPERIENCE

- **2002–2004, 2008, 2013–2014, 2017.** Mentor. *Research Experience for Undergraduates*. Fushan Research Center & Fushan Forest Dynamics Plot, Taiwan.
- **2007–2012.** Teaching Assistant. *Service Education: Herbarium Specimen Archive*, Department of Life Science, National Taiwan University, Taiwan. (seven semesters)
- **2008.** Teaching Assistant. *Service Education: Campus Grasslands Survey*, Department of Life Science, National Taiwan University, Taiwan. (one semester)
- **2008.** Teaching Assistant. *General Biology Lab*. Department of Life Science, National Taiwan University, Taiwan. (one semester)
- **2007.** Teaching Assistant. *Service Education: Environmental Consciousness*, Department of Life Science, National Taiwan University, Taiwan. (one semester)
- **2003.** Teaching Assistant. *Flora and Vegetation of Taiwan*, Department of Botany, National Taiwan University, Taiwan. (one semester)

SERVICE

Peer Reviewer for scientific journals

Applied Vegetation Science (1)
Biotropica (3)
Ecological Research (1)
Journal of Marine Science and Technology (1)
PLoS ONE (2)
Plant Ecology (1)
Population Ecology (1)
Taiwania (3)